2026-10-16 | VS&A Daily |

Before the hammer: pre-test analysis for a small finned rocket



Black Brant IX carrying Dynamo-2 from Wallops, July 2021. A real finned sounding rocket provides context; this is not a modal-test photograph or the worked-example vehicle. Credit: NASA/Allison Stancil. NASA image source [1].

Four accelerometers. A slender airframe. A deceptively easy plan.

Put the sensors along the body, tap sideways, and the bending peaks should appear. But what would that experiment tell you about fin twist, the other bending plane, or a torsional mode with almost no radial motion? A neat instrumentation drawing can leave the most important question unasked.

Today's objective is to turn a pre-test finite element model into a defensible measurement and excitation plan. We will start with the decisions the model must support, build a candidate layout, and use a four-channel fin example to expose a blind spot before hardware reaches the lab.

This advances the October 9 lesson on spatial aliasing into test planning. The layout and numbers are illustrative, not flight-hardware instructions. Published rocket and spacecraft tests provide methodological evidence; none establishes a universal accelerometer count for a small rocket.

Define the experiment before choosing the sensors

First decide what the correlated model must predict: global loads, fin-root stress, aeroelastic behavior, or control-sensitive bending. Set the upper frequency and target list accordingly. Include both lateral bending planes and close pairs, axial and body-torsional modes where relevant, and local fin bending, twist and root coupling. A mode with little total effective mass can still matter greatly to one fin attachment.

Freeze the tested configuration: motor or inert mass simulator, payload, fasteners, joint preload and wiring. Check total mass, center of gravity and inertia against the FEM; verify connectivity, units, mesh convergence and the six free rigid-body modes. Flight and test configurations need separate, traceable definitions.

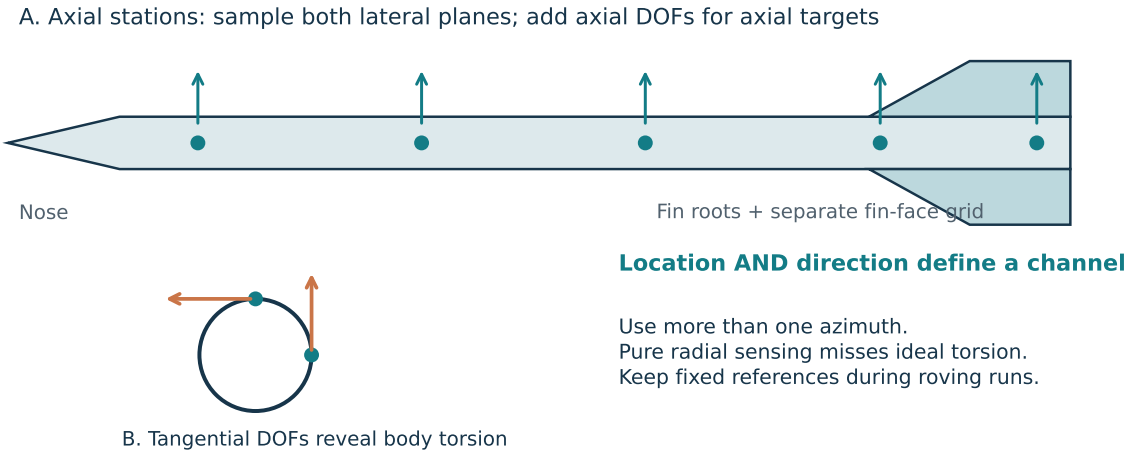


Figure 1. Original candidate-layout concept, not a prescribed channel count. Arrows indicate selected sensing directions; the second bending plane is out of the side-view page. Tangential arrows in the cross-section distinguish torsion from radial motion.

For a nominal free-free survey, represent the suspension rather than assuming it vanishes. KARI’s engine campaign screened rigid-body frequencies against the first elastic mode: below 10% for components and below 20% for larger assemblies. Its complete-engine model included suspension effects when a softer setup was impractical [2]. Treat those ratios as documented screening precedents, not proof of negligible coupling.

Move supports in the model and vary their stiffness. A first-bending node is not a node of every target mode. Instrument supports sufficiently to recognize their motion. For a fixture-mounted test, model the fixture and measure across its interface. Match the experiment you can actually perform.

Begin with generous coverage of load paths, joints and predicted modal motion, then reduce the candidate set. NASA’s ISS Node study used that engineering-judgment-first approach before algorithmic selection [3]. On a small rocket, protect fin and joint coverage when a global metric tries to discard it.

Make the FEM take the proposed test

Export the target eigenvectors at accessible sensor locations. Project each translation onto the installed sensing axis, including interpolation and coordinate transformations. With full-model modes Φ and measurement mapping S, the shapes the instrument can see are:

$$\Psi = S\Phi, \qquad a = \Psi\ddot{q} + \varepsilon$$

A triaxial package provides three channels, but not necessarily three useful directions for every target. Examine each column of Ψ for weak response, and compare columns with AutoMAC for shape ambiguity. Inspect singular values after stating the modal normalization and channel weighting; a small singular value flags a poorly distinguishable combination, even when individual ordinates look large.

Here a is measured acceleration, q contains modal amplitudes and ε is measurement noise. With noise covariance R, a useful information matrix is:

$$F = \Psi^T R^{-1} \Psi$$

F describes information about the chosen modal accelerations. Effective Independence selects coordinates to preserve independence of that modal basis. Full column rank requires at least as many independent rows as retained columns for this algebraic reconstruction problem. That is not a universal minimum sensor count for identifying separated resonances across frequencies or roving runs.

ECSS modal-survey guidance connects measurement directions, linear independence, AutoMAC and test predictions [4]. NASA’s ETM-3 motor survey paired Effective Independence selection with test-load simulations [5]. The transferable lesson is to evaluate the experiment, not just optimize a matrix.

Predict acceleration/force FRFs for feasible inputs, realistic damping ranges and the proposed array. A sideways hammer strike may excite bending well but hardly excite torsion; use tangential or offset forces and fin-normal inputs where needed. An impact hammer is efficient for reconnaissance. Shakers provide controlled sustained excitation and level sweeps, but their stingers, mass and load paths become part of the test.

Repeat the screening over plausible joint stiffness, support properties and sensor masses. Keep fixed references with visibility of the target families during roving measurements, and provide a backup reference where loss of one channel would break the dataset. An optimal nominal layout that fails under modest model error is not robust.

Four channels can miss an entire fin mode

Consider an idealized fin with bending displacement symmetric about mid-chord and twist antisymmetric about a mid-chord elastic axis. All sensors measure normal to the fin. Layout A uses four spanwise points on that axis. Layout B uses near-leading- and trailing-edge pairs at two span stations. These selected shape ordinates demonstrate geometry, not a solved rocket FEM.

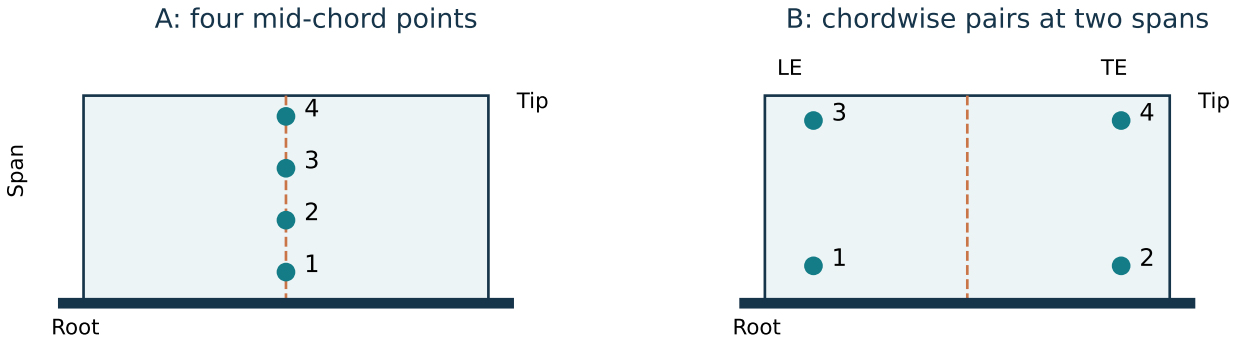


Figure 2. Original fin plan-view sketch. Dashed line: assumed twist axis. LE/TE: leading/trailing edge. Sensors act normal to the page. The simplified rectangular outline is not a flight-fin design.

Channel	A: bending	A: twist	B: bending	B: twist
1	0.2	0	0.2	0.1
2	0.5	0	0.2	-0.1
3	0.8	0	1.0	1.0
4	1.0	0	1.0	-1.0

Table 1. Original dimensionless, consistently scaled illustrative modal ordinates. Four measured DOFs are retained in either layout.

A’s sampled two-mode matrix has rank one: twist is invisible. Its twist AutoMAC is undefined because the twist vector is zero; software must not disguise that as an excellent zero off-diagonal score. B has rank two, and its bending and twist columns have zero dot product.

$$\mathrm{MAC}(b,t)=\frac{|b^{T}t|^{2}}{(b^{T}b)(t^{T}t)}=0 \quad \text{for layout B}$$

The change restores a measurable twist pattern without adding channels. Real fin asymmetry, an offset elastic axis and body motion remove the perfect zeros. That is why you repeat this comparison with actual FEM vectors, credible noise and input participation. Keep root and adjacent-body measurements to distinguish local fin deformation from rigid motion of its attachment.

Use rules of thumb as questions to test

Spatial spacing should resolve the shortest target deformation pattern, not follow a fixed centimetre rule. More than two samples per sinusoidal wavelength avoids the ideal Nyquist ambiguity, but barely meeting that limit does not yield a reliable finite-structure mode shape. Check the actual sampled vectors, including chordwise and circumferential coverage. Large antinode response helps signal quality; it does not ensure independence.

Sensor mass should be judged against local modal inertia, not just total rocket mass. For a small added point mass m_s, negligible added stiffness and an otherwise unchanged mode, first-order eigenvalue perturbation gives:

$$\frac{\Delta f_r}{f_r} \approx -\frac{m_s \phi_r(x_s)^2}{2m_r}$$

Here m_r is generalized modal mass using the same normalization as the local translational ordinate φ_r. For a tip-normalized fin mode with m_r = 20 g, a 1 g sensor at unit ordinate predicts about a 2.5% frequency decrease. This illustrative estimate excludes adhesive, cable stiffness and mode-shape changes; verify by mass perturbation or an alternative measurement. A small fraction of vehicle mass can be a large fraction of fin modal mass.

Frequency resolution follows record duration, not sampling rate alone: $\Delta f = 1/T$. For illustrative modes near 80 Hz separated by 1 Hz and 0.5% damping, the light-damping bandwidth is about 0.8 Hz. An 8 s record gives 0.125 Hz lines, a reasonable starting grid to test, not guaranteed separation. Also verify adequate decay, leakage treatment, anti-alias filtering and usable force spectrum.

A real sounding-rocket case makes the distinction concrete. Faraday Rocketry’s published test used four uniaxial accelerometers and compared two bending frequencies with its FEM. Yet the account explicitly reports poor mode-shape confidence and limited spatial resolution [6]. Frequency agreement alone cannot establish fin-twist observability or readiness for flutter prediction.

The same account describes low suspension frequencies as desirable but labels measured modes near 18-20 Hz as suspension motion, versus first bending near 29 Hz [6]. That discrepancy deserves investigation, not adoption as best practice. The case is valuable evidence of practical limitations; its broad validation language and generic frequency-error tolerance are not universal acceptance criteria.

Leave the planning review with a testable claim

Before the formal survey, check ambient noise, channel polarity and axes, mounting, cable restraint, hammer double hits and force bandwidth. Repeat FRFs, vary force level, and check reciprocity where the structure is linear, passive and reciprocal. Coherence is diagnostic, not a standalone pass/fail number; low values near antiresonances need context. KARI's campaign explicitly used ambient, rigid-body, reciprocity and drive-point checks [2].

Compare frequencies, shapes, FRFs and relevant mass-weighted checks on common measured coordinates. For close pairs, examine the modal subspace rather than forcing a unique vector pairing. FEM-expanded coordinates are inference, not extra independent measurements. Update physically justified joint, mass or stiffness parameters, then check withheld responses. The deliverable is evidence that each target can be excited, observed and distinguished.

Three takeaways

1. Define mode families and test boundaries before counting sensors.
2. Screen both sensing and excitation with the FEM, including uncertainty.
3. Frequency agreement cannot compensate for an unobserved fin or torsional mode.

Reflection. Four mid-chord fin sensors show repeatable bending peaks. Does doubling their sensitivity solve the missing twist measurement?
Answer. No. Sensitivity cannot recover zero geometric participation; change coordinates/directions and verify excitation.

Sources and scope

[1] NASA SVS, Australia Sounding Rocket Campaign Press Kit, released June 7, 2022; Dynamo-2 photograph WFF-2021-045-002, July 2021, Allison Stancil. Photo and credit.

[2] J. Park et al., A Study on the Vibration Modal Testing and Analytical Model Updating of Liquid Rocket Engine of KSLV-II, EUCASS 2019, §§2.2-2.3. Original paper.

[3] M. L. Tinker, Accelerometer Placement for the International Space Station Node Modal Test, AIAA-98-2078, 1998, pp. 1-2. NASA copy.

[4] ECSS-E-ST-32-11C, Modal survey assessment, July 31, 2008, §§5.3 and 6.3. Space-program requirements subject to project applicability and tailoring. Standard.

[5] D. Nielsen et al., Modal Survey of ETM-3, A 5-Segment Derivative of the Space Shuttle Solid Rocket Booster, 2005. Methodological analogue, not a small finned rocket. NASA report.

[6] Dewesoft / Faraday Rocketry, Sounding Rocket Modal Testing for Structural Validation, August 28, 2026; setup, results and limitations. Participant case account.

[7] NASA-STD-5002B, March 10, 2026, §§1.2.1 and 4.9.1. Its pretest mass-orthogonality checks are useful context, but its scope explicitly excludes launch vehicles and sounding-rocket payloads. Do not import its criteria as this rocket's requirements. Standard.

Sources checked October 16, 2026. Figures 1-2, Table 1 and calculations are original illustrations, not measured test data. No proprietary program information is used.